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384. On the Transformation of Plane Curves

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1. The expression a “double point,” or, as I shall for shortness call it, a “dp,” is to be throughout understood to include a cusp: thus, if a curve has δ nodes (or double points in the restricted sense of the expression) and κ cusps, it is here regarded as having $\delta + \kappa$ dps.
2. It was remarked by Cramer, in his “Théorie des Lignes Courbes” (1750), that a curve of the order n has at most $\frac{1}{2}(n-1)(n-2)$, $= \frac{1}{2}(n^2 - 3n) + 1$, dps.
3. For several years past it has further been known that a curve such that the coordinates $(x : y : z)$ of any point thereof are as rational and integral functions of the order n of a variable parameter θ , is a curve of the order n having this maximum number $\frac{1}{2}(n-1)(n-2)$ of dps.
4. The converse theorem is also true, viz.: in a curve of the order n , with $\frac{1}{2}(n-1)(n-2)$ dps, the coordinates $(x : y : z)$ of any point are as rational and integral functions of the order n of a variable parameter θ —or, somewhat less precisely, the coordinates are expressible rationally in terms of a parameter θ .
5. The foregoing theorem, as a particular case of Riemann’s general theorem, to be presently referred to, dates from the year 1857; but it was first explicitly stated only last year (1864) by Clebsch, in the Paper, “Ueber diejenigen ebenen Curven deren Coordinaten rationale Functionen eines Parameters sind,” Crelle, t. LXIV. (1864), pp. 43–63.
6. The demonstration is, in fact, very simple; it depends merely on the remark that we may, through the $\frac{1}{2}(n-1)(n-2)$ dps, and through $2n-3$ other points on the given curve of the order n , together $\frac{1}{2}(n^2+n)-2 = \frac{1}{2}(n-1)(n+2)-1$, points, draw a series of curves of the order $n-1$, given by an equation $U + \theta V = 0$, containing an arbitrary parameter θ ; any such curve intersects the given curve in the dps, each counting as two points, in the $2n-3$ points, and in one other point; hence, as there is only one variable point of intersection, the coordinates of this point, viz., the coordinates of an arbitrary point on the given curve, are expressible rationally in terms of the parameter θ .

The demonstration may also be effected in a similar manner by means of curves of the order $n-2$.

7. Before going further, it will be convenient to introduce the term “Deficiency,” viz., a curve of the order n with $\frac{1}{2}(n-1)(n-2) - D$ dps, is said to have a deficiency $= D$: the foregoing theorem is that for curves with a deficiency $= 0$, the coordinates are expressible rationally in terms of a parameter θ .

Since in such a curve the different points succeed each other in a certain definite order, viz., in the order obtained by giving to the parameter its different real values from $-\infty$ to ∞ , the curve may be termed a *unicursal* curve.

8. Riemann's general theorem, as applied to plane curves, is stated, but not in its complete form, by Schwarz, in the Paper, "De superficiebus in planum explicabilibus primorum septem ordinum," Crelle, t. LXIV. (1864), pp. 1–17: to complete the enunciation it is necessary to refer to page 137 of Riemann's own Paper, "Theorie der Abel'schen Functionen," Crelle, t. LIV. (1857), pp. 115–155, viz., the enunciation will be:
9. For a curve of any order with a given deficiency D , the coordinates may be expressed as follows:

$D = 0$, rationally in terms of a parameter θ , or what comes to the same thing, rationally in terms of the parameters (ξ, η) , connected by an equation of the form $(1, \xi)(1, \eta) = 0$.

$D > 0$, rationally in terms of the parameters (ξ, η) connected by an equation of a certain form, viz.:

$D = 1$, the equation is $(1, \xi)^2(1, \eta)^2 = 0$, or (what comes to the same thing) η is the square root of a quartic function of ξ .

$D = 2$, the equation is $(1, \xi)^2(1, \eta)^3 = 0$, or (what comes to the same thing) η is the square root of a sextic function of ξ .

$D > 2$, viz.:

D odd, $= 2\mu - 3$, the equation is $(1, \xi)^\mu(1, \eta)^\mu = 0$, and is besides such, that treating (ξ, η) as Cartesian coordinates, the curve thereby represented has $(\mu - 2)^2$ dps.

D even, $= 2\mu - 2$, the equation is $(1, \xi)^\mu(1, \eta)^{\mu-1} = 0$, and is besides such, that treating (ξ, η) as Cartesian coordinates, the curve thereby represented has $(\mu - 1)(\mu - 3)$ dps.
10. To see more clearly the meaning of this, write $\frac{\xi}{\zeta}, \frac{\eta}{\zeta}$ in place of (ξ, η) , so that the coordinates $(x : y : z)$ are expressible rationally and homogeneously in terms of (ξ, η, ζ) , connected by an equation of the form $(\xi, \eta, \zeta)^\mu(\xi, \eta, \zeta)^\mu = 0$.

Such an equation, treating therein (ξ, η, ζ) as coordinates, belongs to a curve of the order 2μ , with a μ -tuple point at $(\xi = 0, \zeta = 0)$, a μ -tuple point at $(\eta = 0, \zeta = 0)$, and which has besides $(\mu - 2)^2$ or $(\mu - 1)(\mu - 3)$ dps, according as $D = 2\mu - 3$, or $2\mu - 2$.

The coordinates $(x : y : z)$ of a point of the given curve are expressible rationally in terms of the coordinates $(\xi : \eta : \zeta)$ of a point on the new curve; and we may say that the original curve is by means of the equations which give $(x : y : z)$ in terms of $(\xi : \eta : \zeta)$ transformed into the new curve.
11. A curve of the order 2μ may have $\frac{1}{2}(2\mu - 1)(2\mu - 2) = 2\mu^2 - 3\mu + 1$ dps; hence in the new curve, observing that the μ -tuple points each count for $\frac{1}{2}(\mu^2 - \mu)$ dps, we have

$$\begin{aligned} \text{In the case } D = 2\mu - 3, \quad \text{Deficiency} &= 2\mu^2 - 3\mu + 1 - \mu^2 + \mu - (\mu - 2)^2 \\ &= 2\mu - 3 = D \\ \text{In the case } D = 2\mu - 2, \quad \text{Deficiency} &= 2\mu^2 - 3\mu + 1 - \mu^2 + \mu - (\mu - 1)(\mu - 3) \\ &= 2\mu - 2 = D \end{aligned}$$

Moreover for $D = 0$, the transformed curve is a conic, with 0 dps, and therefore with deficiency = 0; in the case $D = 1$, it is a quartic with 2 dps, and therefore deficiency = 2; in the case $D = 2$, it is a quintic with a triple point = 3, and a double point = 1, together 4 dps, and therefore deficiency = 2.

Hence in every case the new curve has the same deficiency as the original curve.

12. The theorem thus is that the given curve of the order n , with deficiency D , may be rationally transformed into a curve of an order depending only on the deficiency, and having the same deficiency with the given curve, viz.:

$D = 0$, the new curve is of the order $2(= D + 2)$; $D = 1$, it is of the order $4(= D + 3)$; $D = 2$, it is of the order $5(= D + 3)$; and $D > 2$, it is for D odd, of the order $D + 3$; and for D even, of the order $D + 2$.

It will presently appear that these are not the lowest values which it is possible to give to the order of the new curve.

Riemann's object was, not that the order of the transformed curve might be as low as possible, but that the equation in (ξ, η) might be in each of these parameters separately of the lowest possible order; and this he effected by giving to the transformed curve the two μ -tuple points.

13. It is to be noticed that the theorem that for any rational transformation of one curve into another the two curves have the same deficiency is in effect given (as a consequence of Riemann's theory) by Clebsch in the Paper, "Ueber die Singularitäten algebraischer Curven," Crelle, t. LXIV., pp. 98–100. I have, by the assistance of a formula communicated to me by Dr Salmon, obtained a direct analytical demonstration of this theorem.
14. I remark that (x, y, z) being connected by an equation, if $(x : y : z)$ are given rationally in terms of $(\xi : \eta : \zeta)$, then it follows that $(\xi : \eta : \zeta)$ are also expressible rationally in terms of $(x : y : z)$: it is convenient to consider the transformation from this point of view, and I now proceed to the independent development of the theory, as follows:
15. We have a given curve $U = (x, y, z)^n = 0$, with deficiency D , which is by the transformation $\xi : \eta : \zeta = P : Q : R$ (where P, Q, R are given functions $(x, y, z)^k$ each of the same order k) transformed into the curve $T = (\xi, \eta, \zeta)^{n'} = 0$. The transformed curve has, as we know, the same deficiency D as the original curve.
16. To find the order of the transformed curve, we must find the number of its intersections with an arbitrary line $a\xi + b\eta + c\zeta = 0$. Writing in this equation $\xi : \eta : \zeta = P : Q : R$, we obtain the equation $aP + bQ + cR = 0$, and combining therewith the equation $U = 0$, the two equations, being of the orders k and n respectively, give kn systems of values of $(x : y : z)$, and to each of these, in virtue of the equations $\xi : \eta : \zeta = P : Q : R$, there corresponds a single set of values of $(\xi : \eta : \zeta)$, and therefore a single point of intersection; hence the number of intersections, that is, the order of the transformed curve, is $= kn$.
17. If, however, the curves $P = 0, Q = 0, R = 0$, meet in an ordinary point of the curve $U = 0$, then it is easy to see there is a reduction $= 1$ in the foregoing value; and so if they meet in a dp of the curve $U = 0$, then there is a reduction $= 2$. More generally if the curves $P = 0, Q = 0, R = 0$ each pass through the same α dps and β ordinary points of the curve $U = 0$, then there is a reduction $= 2\alpha + \beta$.
- In fact the curve $aP + bQ + cR = 0$, meets the curve $U = 0$, in kn points; but among these are included the α dps, each counting as 2 intersections, and the β points; the number of the remaining intersections is $= kn - 2\alpha - \beta$, and the order of the transformed curve is equal to this number. I assume that we have $k < n$.
18. A curve of the order k may be made to pass through $\frac{1}{2}k(k + 3)$ points; it is moreover known that if any three curves, $P = 0, Q = 0, R = 0$, of the order k each pass through the same $\frac{1}{2}k(k + 3) - 1$ points, then the three curves have all their intersections common, the equations being, in fact, connected by an identical relation of the form $\alpha P + \beta Q + \gamma R = 0$.

To make the order of the transformed curve as low as possible, we must make the curves $P = 0$, $Q = 0$, $R = 0$ meet on the curve $U = 0$ in as many points as possible, and it appears from the remark just made, that the greatest possible number is $= \frac{1}{2}k(k+3) - 1$; and we have thus for the order of the transformed curve

$$n' = kn - 2\alpha - \beta,$$

where α, β are any numbers satisfying the equation

$$2\alpha + \beta = \frac{1}{2}k(k+3) - 1.$$

But we must have $n' > n$ (for otherwise the transformation is illusory, the transformed curve being transformable into the original curve by a transformation of the same or a lower order); and hence we must have

$$kn - \frac{1}{2}k(k+3) + 1 > n,$$

or, what is the same thing,

$$(k-1)(n - \frac{1}{2}k - 2) > 1,$$

which implies $k \geq 2$, $n \geq \frac{1}{2}k + 3$; that is, $n \geq 4$ for $k = 2$, and $n \geq 5$ for $k = 3$, etc.

19. I consider in particular $k = 2$, that is, the transformation by means of conics through $\frac{1}{2} \cdot 2 \cdot 5 - 1 = 4$ points on the curve. Here

$$n' = 2n - 2\alpha - \beta,$$

where $2\alpha + \beta = 4$; so that the pairs of values are $(\alpha, \beta) = (2, 0)$, $(1, 2)$, or $(0, 4)$.

The case $(\alpha, \beta) = (2, 0)$ gives $n' = 2n - 4$; that is, a curve of the order n with 2 dps is transformed into a curve of the order $2n - 4$ with the same deficiency. This deficiency $D = \frac{1}{2}(n-1)(n-2) - 2 = \frac{1}{2}(n^2 - 3n)$; and correspondingly $n' = 2n - 4$ gives $D = \frac{1}{2}(2n-5)(2n-6) - \frac{1}{2}(2n-4)(2n-5) + 2(2n-5) = \frac{1}{2}(n^2 - 3n)$ as before.

20. I stop here to remark that the foregoing transformation gives rise to a transformation of the most general nature. Supposing that the curves $P = 0$, $Q = 0$, $R = 0$ each pass through the same points on the given curve $U = 0$, then if we consider the series of curves $\Pi = P + \lambda Q + \mu R = 0$, any such curve meets the given curve in the fixed points (each counting as two intersections for a dp) and in one variable point; the coordinates of this point are rationally expressible in terms of (λ, μ) ; that is, we have $x : y : z = \phi(\lambda, \mu) : \psi(\lambda, \mu) : \chi(\lambda, \mu)$, where ϕ, ψ, χ are rational functions of (λ, μ) . Conversely, given any rational expression of the coordinates in terms of two parameters, we may by elimination of the parameters obtain the equation of the curve; and the transformation may be regarded as establishing a $(1, 1)$ correspondence between the points of the given curve and the points of a plane (λ, μ) .
21. We may consider the question of reducing the order of the transformed curve by reference to the theory of linear series. If the given curve $U = 0$ has deficiency D , then the complete series of groups of m points cut out on the curve by curves of any given order is of the dimension $m - D$ (when $m \geq D$); and for a general curve of order n , a special series does not exist. But for a curve with dps, or more generally for any curve of deficiency D , there may exist special series; and it is by means of these that the order of the transformed curve may be reduced below the value given by Riemann's transformation.

22. In particular, for a curve with deficiency D , there exists a g_{D+2}^1 (a linear series of dimension 1, degree $D + 2$); that is, the curve may be transformed into a curve of order $D + 2$. For $D = 0$, this is a conic (order 2); for $D = 1$, a cubic (order 3); for $D = 2$, a quartic with one dp (order 4); and so on. These are in fact the lowest possible orders.
23. To verify that the transformation is indeed general, I observe that for a general curve of order n with deficiency D , the number of absolute constants (moduli) is $3D - 3$ (for $D \geq 2$); and the transformation depends on a certain number of parameters which must be at least equal to this, in order that the transformation may be general. The number of conditions to be satisfied is precisely the number required to determine the moduli, which confirms that the transformation is indeed general.
24. I remark in conclusion that the transformation theory here developed has important applications to the theory of integrals belonging to the curve; in particular, the Abelian integrals of the first kind belonging to the given curve correspond to similar integrals belonging to the transformed curve; and this correspondence furnishes a powerful method for investigating the properties of such integrals.

385. On the Correspondence of Two Points on a Curve

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1. In a unicursal curve the coordinates (x, y, z) of any point of the curve are proportional to rational and integral functions of a variable parameter θ . Hence, if two points of the curve correspond in suchwise that to a given position of the first point there correspond α' positions of the second point, and to a given position of the second point α positions of the first point, the number of points which correspond each to itself is $= \alpha + \alpha'$.

For let the two points be determined by their parameters θ, θ' respectively—then to a given value of θ there correspond α' values of θ' , and to a given value of θ' there correspond α values of θ ; hence the relation between (θ, θ') is of the form $(\theta, 1)^{\alpha'}(\theta', 1)^{\alpha} = 0$; and writing therein $\theta' = \theta$, then for the points which correspond each to itself, we have an equation $(\theta, 1)^{\alpha+\alpha'} = 0$ of the order $\alpha + \alpha'$; that is, the number of these points is $= \alpha + \alpha'$.

2. Hence for a unicursal curve we have a theorem similar to that of M. Chasles' for a line, viz., the theorem may be thus stated:

If two points of a unicursal curve have an (α, α') correspondence, the number of united points is $= \alpha + \alpha'$.

3. But a unicursal curve is nothing else than a curve with a deficiency $D = 0$, and we thence infer

Theorem 1. *If two points of a curve with deficiency D have an (α, α') correspondence, the number of united points is $= \alpha + \alpha' + 2kD$; in which theorem $2k$ is a coefficient to be determined.*

4. Suppose that the corresponding points are P, P' and imagine that when P is given, the corresponding points P' are the intersections of the given curve by a curve Θ (the equation of the curve will of course contain the coordinates of P as parameters, for otherwise the position of P' would not depend upon that of P).

I find that if the curve Θ has with the given curve k intersections at the point P , then in the system of (P, P') , the number of united points is

$$\sigma = \alpha + \alpha' + 2kD,$$

whence in particular, if the curve Θ does not pass through the point P , then the number of united points is $= \alpha + \alpha'$, as in a unicursal curve.

- 4*. The foregoing theorem is easily proved in the particular case where the k intersections at the point P take place in consequence of the curve Θ having a k -tuple point at P .

Taking $U = (x, y, z)^m = 0$ as the equation of the given curve (which for greater simplicity is assumed to be a curve without singularities), then if we suppose (x, y, z) to be the coordinates of the point P , and (x', y', z') to be the coordinates of the point P' , write

$$U = (x, y, z)^m, \quad U' = (x', y', z')^m,$$

U' being what U becomes on writing therein (x', y', z') in place of (x, y, z) ; and

$$\Theta = (x, y, z)^k (x', y', z')^{k'} (yz' - y'z, zx' - z'x, xy' - x'y)^k,$$

viz., Θ is a function of the order k in $yz' - y'z, zx' - z'x, xy' - x'y$, the coefficients of the several powers and products of these quantities being functions of the order α in (x, y, z) and of the order α' in (x', y', z') , which functions are such that they do not all of them vanish identically, or in virtue of the equation $U = 0$, on writing therein $(x', y', z') = (x, y, z)$.

Taking for a moment (x, y, z) as current coordinates, suppose that the equation of the given curve is $U = 0$; then if (x, y, z) are the coordinates of the point P , we have $U = 0$, and similarly if (x', y', z') are the coordinates of the point P' we have $U' = 0$.

The equation $\Theta = 0$, considering therein (x, y, z) as the coordinates of the given point P (and so as parameters satisfying the equation $U = 0$) and (x', y', z') as current coordinates, will be a curve having a k -tuple point at P , we have thus the case above supposed; and P being given, the corresponding points P' are given as the intersections of the curves $U' = 0, \Theta = 0$, which are respectively of the orders m and $\alpha' + k$; the total number of intersections is thus $= m(\alpha' + k)$, but inasmuch as the curve $\Theta = 0$ has a k -tuple point at P , k of these intersections coincide with the point P , and the number of the remaining intersections, that is the number of positions of the point P' , is $= m\alpha' + (m - 1)k$.

Similarly when P' is given, the number of positions of the point P is $= m\alpha + (m - 1)k$: and we have therefore

$$\alpha + \alpha' = m(\alpha + \alpha') + 2(m - 1)k.$$

To find the united points, it is to be observed, that upon writing $(x', y', z') = (x, y, z)$, the function Θ becomes identically $= 0$; but if we suppose, in the first instance, that P, P' are consecutive points on the curve $U = 0$, then we have

$$yz' - y'z : zx' - z'x : xy' - x'y = \partial_x U : \partial_y U : \partial_z U;$$

and the equation $\Theta = 0$ assumes the form

$$\Theta = (x, y, z)^k (x, y, z)^k (\partial_x U, \partial_y U, \partial_z U)^k = 0,$$

which, (x, y, z) being current coordinates, is the equation of a curve of the order $\alpha + \alpha' + (m - 1)k$; the united points are the intersection of this curve with the given curve $U = 0$, and the number of the united points is thus

$$\sigma = m(\alpha + \alpha') + m(m - 1)k.$$

Hence attending to the above-mentioned value of $\alpha + \alpha'$, we have

$$\sigma = \alpha + \alpha' + (m-1)(m-2)k.$$

But in the case of a curve $U = 0$, without singularities, we have

$$2D = (m-1)(m-2),$$

and we have thus the required formula

$$\sigma = \alpha + \alpha' + 2kD.$$

The investigation in the case where the k intersections at P arise wholly or in part from a contact of the curve Θ , or any branch or branches thereof, with the given curve U , is more difficult, and I abstain from entering upon it.

I apply the theorem to some examples:

5. Investigation of the class of a curve of the order m with δ dps.

Take as corresponding points on the curve two points, such that the line joining them passes through a fixed point O : the united points will be the points of contact of the tangents through O ; that is, the number of the united points is equal to the class of the curve.

The curve Θ is here the line OP , which has with the given curve a single intersection at P ; that is, we have $k = 1$.

The points P' corresponding to a given position of P are the remaining $m-1$ intersections of OP with the curve; that is, we have $\alpha' = m-1$; and in like manner $\alpha = m-1$.

Hence the class is $= 2(m-1) + 2D$, viz., this is $= (2m-2) + (m^2 - 3m + 2 - 2\delta)$, which is $= m^2 - m - 2\delta$, as it should be.

6. It is in the foregoing example assumed that the dps are none of them cusps; if the curve has $\delta + \kappa$ dps, κ of which are cusps (or what is the same thing, δ nodes and κ cusps); then the number of united points is equal $2(m-1) + 2D$, $= m^2 - m - 2\delta - 2\kappa$; but in this case each of the cusps is reckoned as a united point, and we have, therefore,

$$\text{class} + \kappa = m^2 - m - 2\delta - 2\kappa,$$

that is,

$$\text{class} = m^2 - m - 2\delta - 3\kappa.$$

This will serve as an instance of the special considerations which are required in the case of a curve with cusps, but in what follows, I shall assume that the dps are none of them cusps and thus attend to the case of a curve of the order m , with δ dps, and therefore of the class $n = m^2 - m - 2\delta$, and of the deficiency

$$D = \frac{1}{2}(m-1)(m-2) - \delta = \frac{1}{2}(n - 2m + 2).$$

7. Investigation of the number of inflexions.

Taking the point P' to be a tangential of P (that is, an intersection of the curve by the tangent at P), the united points are the inflexions, and the number of the united points is equal to the number of inflexions.

The curve Θ is here the tangent at P , having with the given curve two intersections at P ; that is $k = 2$.

P' is any one of the $m - 2$ tangentials of P , hence $\alpha' = m - 2$; and P is the point of contact of any one of the $m - 2$ tangents from P' to the curve, that is, $\alpha = m - 2$.

Hence the number of inflexions is

$$= (m - 2) + (m - 2) + 4D = m + n - 4 + 2(n - 2m + 2) = 3(n - m),$$

which is right.

8. For the purpose of the next example it is necessary to present the fundamental equation $\sigma = \alpha + \alpha' + 2kD$ under a more general form.

The curve Θ may intersect the given curve in a system of points P' each p times, a system of points Q' each q times, &c., in such manner that the points (P, P') , the points (P, Q') , &c., are pairs of points corresponding to each other according to distinct laws; and we shall then have the numbers $(\alpha, \alpha, \alpha')$, (β, β, β') , &c., belonging to these pairs respectively; viz. (P, P') are points having an (α, α') correspondence, and the number of united points is $= a$; similarly (P, Q') are points having a (β, β') correspondence, and the number of united points is $= b$; and so on.

The theorem then is

$$p(a - \alpha - \alpha') + q(b - \beta - \beta') + \cdots = 0,$$

or, as this may also be written,

$$pa + qb + \cdots = p(\alpha + \alpha') + q(\beta + \beta') + \cdots + 2(p\alpha + q\beta + \cdots)D,$$

where $p\alpha + q\beta + \cdots = k$ (a positive integer).

9. As an application of the generalized formula, consider the correspondence of two points P, P' on a curve such that the line PP' meets a given curve of the order r in k points (besides the points P, P'). The united points are the points of contact of the tangents from the points of intersection of the given curve $U = 0$ with the curve of the order r ; or, what is the same thing, the points of the curve $U = 0$ which lie on the double tangents of this curve which meet the curve of the order r .

We have here a single class of corresponding points, with $p = 1$, $\alpha = \alpha' = m - 1$, $a = n - m + 2D$ (the number of tangents from a point of the curve to the curve itself); hence

$$n - m + 2D - 2(m - 1) = 2kD,$$

that is,

$$n - 3m + 2 = 2(k - 1)D,$$

or, what is the same thing, $k = 1$ gives $n = 3m - 2 - 2D = m + 2\delta$; that is $n = m^2 - m - 2\delta$.

10. The theorem of correspondence may be further extended to the case of two curves with different deficiencies D, D' , but I do not pursue this extension here.

386. On the Logarithms of Imaginary Quantities

[From the Proceedings of the London Mathematical Society, vol. II. (1866–1869), pp. 50–54.]

1. The logarithm of a real positive quantity is determinate; but for a real negative quantity, or for an imaginary quantity, the logarithm has in the first instance an arbitrary value; and it is only when a specific value has been selected that the logarithm becomes a determinate function of the quantity, capable of being operated upon in the ordinary manner. The selected value is taken to be the one which is continuous with the real value for a positive quantity, as this quantity passes continuously from positive to negative through a value zero; or, what is the same thing, for a quantity $x + iy$, where x is positive, the selected value is the ordinary real value $\log \sqrt{x^2 + y^2} + i \arctan \frac{y}{x}$, and for other values of x the selected value is determined by the condition of continuity. But inasmuch as when the point (x, y) describes a closed curve round the origin, the value of the logarithm is increased by $2i\pi$, it follows that the selected value has a line of discontinuity extending from the origin to infinity; this line is usually taken to be the negative part of the axis of x .
2. Let P be any point (x, y) , then taking the selected value, we have

$$\log P = \log r + i\theta,$$

where $r = \sqrt{x^2 + y^2}$ is the distance from the origin, and θ is the angle measured from the positive axis of x to the radius vector OP ; the value of θ being taken between the limits $-\pi$ and π . It may be remarked that the value $\log P'' - \log P'$ of the definite integral $\int_{P'}^{P''} \frac{dz}{z}$, taken along any path not crossing the line of discontinuity, is equal to $\log P'' - \log P'$, where these logarithms are the selected values at the points P'' and P' respectively.

3. If however the path of integration from P' to P'' crosses the line of discontinuity, then the value of the definite integral is not $= \log P'' - \log P'$, but it differs from this by $\pm 2i\pi$. In fact, if the chord $P'P''$ cuts the negative part of the axis of x upwards, there is an abrupt change from $-i\pi$ to $+i\pi$; if the chord $P'P''$ cuts the negative part of the axis of x downwards, there is an abrupt change from $+i\pi$ to $-i\pi$; in the former case, by taking the definite integral to be $\log P'' - \log P'$, we take its value too large by $2i\pi$, in the latter case we take it too small by $2i\pi$; that is, the true value of the definite integral is in the former case $= \log P'' - \log P' - 2i\pi$, in the latter case it is $= \log P'' - \log P' + 2i\pi$.

But if the chord $P'P''$ does not cut the negative part of the axis of x , then there is not any discontinuity, and the true value of the definite integral is $= \log P'' - \log P'$.

We have thus in the three cases respectively

$$\log p = \log P'' - \log P', \quad p = \log P'' - \log P' + 2i\pi, \quad p = \log P'' - \log P' - 2i\pi,$$

which agrees with the previous results.

4. It may be remarked, that it is merely in consequence of the particular definition adopted that there is in the value of $\log P$ a discontinuity at the passage over the negative part of the axis of x ; with a different definition of the logarithm, there would be a discontinuity at the passage over some other line from the origin; but a discontinuity somewhere there must be.

For if, as above, the chord $P'P$ meet the negative part of the axis of x , then forming a closed quadrilateral by joining by right lines the points 1 to P , P to P' , P' to $\frac{1}{p}$, and $\frac{1}{p}$ to 1; the only side meeting the negative part of the axis of x is the side $P'P$; the integral $\int \frac{dz}{z}$, taken through the closed circuit in question, or say the integral

$$\int_1^P \frac{dz}{z} + \int_P^{P'} \frac{dz}{z} + \int_{P'}^{1/p} \frac{dz}{z} + \int_{1/p}^1 \frac{dz}{z}$$

has, by what precedes, a value in consequence of the discontinuity in passing from P' to P ; viz., this is $= -2i\pi$ or $= +2i\pi$, according as the chord $P'P$, considered as drawn from P' to P , cuts the negative part of the axis of x upwards or downwards; but this value $-2i\pi$ or $+2i\pi$ must be altogether independent of the definition of the logarithm; whereas if, by any alteration in the definition, the discontinuity could be avoided, the value of the integral, instead of being as above, would be $= 0$.

The foregoing value $-2i\pi$ or $+2i\pi$ is in fact that of the integral taken along (in the one or the other direction) any closed curve surrounding the point $z = 0$, for which the function $\frac{1}{z}$ under the integral sign becomes infinite: but in obtaining the value as above, no use is made of the principles relating to the integration of functions which thus become infinite.

5. The equation $\log P = \log r + i\theta$ gives

$$(x + iy)^m = r^m e^{im\theta},$$

or say

$$(x + iy)^m = r^m e^{im\theta},$$

where, m being any real quantity whatever, r^m denotes the positive real value of r^m .

We have thus a definition of the value of $(x + iy)^m$, and the value so defined may be called the *selected value*.

And similarly, for an imaginary exponent $m = p + qi$, we have

$$\begin{aligned} (x + iy)^{p+qi} &= e^{(p+qi)(\log r + i\theta)} \\ &= e^{p \log r - q\theta + i(p\theta + q \log r)} \\ &= r^p e^{-q\theta} \cdot e^{i(p\theta + q \log r)} \end{aligned}$$

which is the selected value of $(x + iy)^{p+qi}$.

6. It may be remarked, in illustration of the advantage (or rather the necessity) of having a selected value, that in an integral $\int Z dz$, taken between given limits along a given path, it is necessary that we know, for the real or imaginary value of z corresponding to each point of the path, the value of the function Z ; and consequently, if Z is a function involving $\log z$ or z^m , the indeterminateness which presents itself in these symbols (considered as belonging to a single value of z) is, so to speak, indefinitely multiplied, and $\int Z dz$ is really an unmeaning combination of symbols, unless by selecting, as above or otherwise, a unique value of $\log z$ or z^m , we render the function to be integrated a determinate function of the variable.

387. Notices of Communications to the London Mathematical Society

[From the Proceedings of the London Mathematical Society, vol. II. (1866–1869),
pp. 6–7, 25–26, 29, 61–63, 103–104, 123–125.]

December 13, 1866. pp. 6–7

Prof. Cayley exhibited and explained some geometrical drawings. Thinking that the information might be convenient for persons wishing to make similar drawings, he noticed that the paper used was a tinted drawing paper, made in continuous lengths up to 24 yards, and of the breadth

of about 56 inches; the half-breadth being therefore sufficient for ordinary figures, and the paper being of a good quality and taking colour very readily.

Among the drawings was one of the conics through four points forming a convex quadrangle. The plane is here divided into regions by the lines joining each of the six pairs of points, and by the two parabolas through the four points; and the regions being distinguished by different colours, the general form of the conics of the system is very clearly seen.

(Prof. Cayley remarked that it would be interesting to make the figures of other systems of conics satisfying four conditions; and in particular for the remaining elementary systems of conics, where the conics pass through a number 3, 2, 1 or 0 of points and touch a number 1, 2, 3 or 4 of lines: the construction of some of these figures is, however, practically a great deal more difficult.)

Other figures related to Cartesians and Bicircular Quartics. One of these was a figure of a system of triconfocal Cartesians; and derived from this by inversion in regard to a circle, there was a figure of a system of quadricconfocal bicircular quartics: in the assumed position of the inverting circle, each quartic consists (like the Cartesian which gives rise to it) of an exterior and an interior continuous curve, and the general aspect of the figure is that of a distortion of the original figure of the Cartesians.

Another figure was that of the bicircular quartic, for which the algebraical sum of the distances of a point thereof from three given foci is $= 0$ (this was selected for facility of construction, by the intersections of circles and confocal conics). The quartic consists of two equal and symmetrically situated pear-shaped curves, exterior to each other, and including the one of them two of the three given foci, the other of them the third given focus, and a fourth focus lying in a circle with the given foci: by inversion in regard to a circle having its centre at a focus the two pear-shaped curves became respectively the exterior and the interior ovals of a Cartesian.

There was also a figure of the two circular cubics, having for foci four given points on a circle; and a figure (coloured in regions) in preparation for the construction of the analogous sextic curve derived from four given points not in a circle.

March 28, 1867. pp. 25–26

Professor Cayley mentioned a theorem included in Prof. Sylvester's theory of derivation of the points of a cubic curve.

Writing down the series of numbers 1, 2, 4, 5, 7, 8, 10, 11, 13, 14, 16, 17, &c., viz., all the numbers not divisible by 3, then (repetitions of the same number being permissible) taking any two numbers of the series, we have in the series a third number, which is the sum or else the difference of the two numbers (for example, 2, 2 give their sum 4, but 2, 7 give their difference 5), and we have thus a series of triads, in each of which one number is the sum of the other two.

The theorem is, that it is possible on a cubic curve to construct a series of points, such that denoting them by the above numbers respectively, then for any triad of numbers as aforesaid the points denoted by the three numbers respectively lie in lines.

And the theorem gives its own construction: in fact the series of triads is 112, 224, 145, 257, 178, 248, &c.

Take 1, an arbitrary point on the cubic, then (by the theorem) the triad 112 shows that 2 is the tangential of 1; 224 shows that 4 is the tangential of 2; 145 that 5 is the third point of 1 and 4; 257 that 7 is the third point of 2 and 5.

So far we have no theorem; we have merely, starting from the point 1, constructed by an arbitrary process the points 2, 4, 5, and 7. But going a step further; 178 and 448 show, the first of them, that 8 is the third point of 1 and 7, the second of them, that 8 is the tangential of 4. We have here the theorem that the third point of 1 and 7 is also the tangential of 4.

Similarly, 10, 11, 13 are each of them (like 8) determined by two constructions; 14, 16, 17, 19, each of them by three constructions, and so on; the number of constructions increasing by unity

for each group of four numbers.

And the theorem is, that these constructions, 2, 3, or more, as the case may be, give always one and the same point.

Prof. Cayley mentioned that on a large figure of a cubic curve he had, in accordance with the theorem, constructed the series of points 1, 2, 4, 5, 7, 8, 10, 11, 13, 14.

April 15, 1867. p. 29

Prof. Cayley communicated a theorem relating to the locus of the ninth of the points of intersection of two cubics, seven of these points being fixed, while the eighth moves on a straight line.

March 26, 1868. pp. 61–63

Prof. Cayley made some remarks on a mode of generation of a sibi-reciprocal surface, that is, a surface the reciprocal of which is of the same order and has the same singularities as the original surface.

If a surface be considered as the envelope of a plane varying according to given conditions, this is a mode of generation which is essentially not sibi-reciprocal; the reciprocal surface is given as the locus of a point varying according to the reciprocal conditions.

But if a surface be considered as the envelope of a quadric surface varying according to given conditions, then the reciprocal surface is given as the envelope of a quadric surface varying according to the reciprocal conditions; and if the conditions be sibi-reciprocal, it follows that the surface is a sibi-reciprocal surface.

For instance, considering the surface which is the envelope of a quadric surface touching each of 8 given lines; the reciprocal surface is here the envelope of a quadric surface touching each of 8 given lines; the conditions are thus sibi-reciprocal, and the surface is therefore a sibi-reciprocal surface.

April 16, 1868. pp. 103–104

Prof. Cayley gave an account of a theorem relating to the sixteen nodes of a quartic surface. The theorem is that for a quartic surface with 16 nodes, the nodes may be regarded as lying in 16 different planes, each plane containing 6 nodes; and that the 16 planes form a symmetrical configuration.

May 14, 1868. pp. 123–125

Prof. Cayley communicated a theorem on the bitangents of a quartic curve. The theorem states that the 28 bitangents of a quartic curve may be divided into 7 sets of 4 bitangents, such that the 8 points of contact of each set of 4 bitangents lie on a conic.

388. Note on the Composition of Infinitesimal Rotations

[From the Quarterly Journal of Pure and Applied Mathematics, vol. VIII. (1867), pp. 7–10.]

1. I call to mind two well-known lemmas:

Lemma 1. The displacements of a point (x, y, z) produced by an infinitesimal rotation ω about an axis having for its six coordinates (a, b, c, f, g, h) , are $\omega(-cy + bz + f)$, $\omega(cx - az + g)$, $\omega(-bx + ay + h)$, in the directions of the axes respectively.

Lemma 2. The displacement of the point (x, y, z) in the direction of a line (α, β, γ) , produced by any infinitesimal motion whatever of a solid body, is $= a\delta x + b\delta y + c\delta z + f\delta p + g\delta q + h\delta r$, where (a, b, c, f, g, h) are the six coordinates of the infinitesimal motion, and $\delta x, \delta y, \delta z, \delta p, \delta q, \delta r$ are the variations of the six coordinates which correspond to the infinitesimal displacement of the point (x, y, z) along the line (α, β, γ) .

2. The first of these lemmas establishes a correspondence between infinitesimal rotations and forces; viz., an infinitesimal rotation ω about the line (a, b, c, f, g, h) corresponds to a force ω along the line (a, b, c, f, g, h) . And it is to be remarked that the correspondence is such that the system of displacements produced by the rotation is the same as the system of displacements which would be produced by the corresponding force, if the points of application of the forces were displaced in directions perpendicular to the axes of rotation, with magnitudes proportional to the distances from these axes.
3. The second lemma shows that the displacement of a point in a given direction, produced by any infinitesimal motion of a solid body, is the same as the sum of the displacements produced by the several rotations into which the motion may be resolved. Hence, by the principle of virtual velocities, if a system of rotations $\omega_1, \omega_2, \&c.$, about the lines $(a_1, b_1, c_1, f_1, g_1, h_1), (a_2, b_2, c_2, f_2, g_2, h_2), \&c.$, are in equilibrium, the displacements $(\delta x, \delta y, \delta z)$ of any point (x, y, z) whatever must each of them vanish; that is, we must have

$$\Sigma \omega a = 0, \quad \Sigma \omega b = 0, \quad \Sigma \omega c = 0, \quad \Sigma \omega f = 0, \quad \Sigma \omega g = 0, \quad \Sigma \omega h = 0,$$

which are therefore the conditions for the equilibrium of the rotations $\omega_1, \omega_2, \&c.$

Secondly, for a system of forces P_1 along the line $(a_1, b_1, c_1, f_1, g_1, h_1), \&c.$, the condition of equilibrium as given by the principle of virtual velocities is

$$\Sigma P(a\delta x + b\delta y + c\delta z + f\delta p + g\delta q + h\delta r) = 0;$$

or, what is the same thing, we must have

$$\Sigma Pa = 0, \quad \Sigma Pb = 0, \quad \Sigma Pc = 0, \quad \Sigma Pf = 0, \quad \Sigma Pg = 0, \quad \Sigma Ph = 0,$$

which are therefore the conditions for the equilibrium of the forces $P_1, P_2, \&c.$

Comparing the two results we see that the conditions for the equilibrium of the rotations $\omega_1, \omega_2, \&c.$, are the same as those for the equilibrium of the forces $P_1, P_2, \&c.$; and since, for rotations and forces respectively, we pass at once from the theory of equilibrium to that of composition; the rules of composition are the same in each case.

4. Demonstration of Lemma 1.

Assuming for a moment that the axis of rotation passes through the origin, then for the point P , coordinates (x, y, z) , the square of the perpendicular distance from the axis is

$$= (-y \cos \gamma + z \cos \beta)^2 + (x \cos \gamma - z \cos \alpha)^2 + (-x \cos \beta + y \cos \alpha)^2,$$

and the expressions which enter into this formula denote as follows; viz. if through the point P , at right angles to the plane through P and the axis of rotation, we draw a line PQ , = perpendicular distance of P from the axis of rotation, then the coordinates of Q referred to P as origin are

$$-y \cos \gamma + z \cos \beta, \quad x \cos \gamma - z \cos \alpha, \quad -x \cos \beta + y \cos \alpha,$$

respectively.

Hence the foregoing quantities each multiplied by ω are the displacements of the point P in the directions of the axes, produced by the rotation ω .

Suppose that the axis of rotation (instead of passing through the origin) passes through the point (x_0, y_0, z_0) ; the only difference is that we must in the formula write $(x - x_0, y - y_0, z - z_0)$ in place of (x, y, z) : and attending to the significations of the six coordinates (a, b, c, f, g, h) it thus appears that the displacements produced by the rotation are equal to ω into the expressions

$$-cy + bz + f, \quad cx - az + g, \quad -bx + ay + h,$$

respectively.

5. Demonstration of Lemma 2.

For any infinitesimal motion whatever of a solid body, the displacements of the point (x, y, z) in the directions of the axes are

$$\delta x = l - ry + qz, \quad \delta y = m + rx - pz, \quad \delta z = n - qx + py,$$

and hence the displacement in the direction of the line (α, β, γ) , is

$$\delta x \cos \alpha + \delta y \cos \beta + \delta z \cos \gamma,$$

which, attending to the signification of the six coordinates (a, b, c, f, g, h) , is

$$= al + bm + cn + fp + gq + hr,$$

which is the required expression.

It is proper to remark that the last-mentioned expressions of $(\delta x, \delta y, \delta z)$ are in fact the displacements produced by a translation and a rotation. If we assume that every infinitesimal motion of a solid body can be resolved into a translation and a rotation, then, since a translation can be produced by two rotations, every infinitesimal motion of a solid body can be resolved into rotations alone, and the foregoing expressions for the displacements produced by a rotation, combining any number of them and writing $(\Sigma \omega a, \Sigma \omega b, \Sigma \omega c, \Sigma \omega f, \Sigma \omega g, \Sigma \omega h) = (-l, -m, -n, l, m, n)$ respectively, lead to the expressions for the displacements $\delta x, \delta y, \delta z$ produced by the infinitesimal motion of the solid body.

389. On a Locus derived from Two Conics

[From the Quarterly Journal of Pure and Applied Mathematics, vol. VIII. (1867), pp. 77-84.]

Required the locus of a point which is such that the pencil formed by the tangents through it to two given conics has a given anharmonic ratio.

1. Suppose, for a moment, that the equation of the tangents to the first conic is $(x - ay)(x - by) = 0$, and that of the tangents to the second conic is $(x - cy)(x - dy) = 0$, and write

$$A = (a - b)(c - d), \quad B = (a - c)(d - b), \quad C = (a - d)(b - c),$$

so that $A + B + C = 0$, write also

$$k = \frac{A'}{A}, \quad k' = \frac{B'}{B}, \quad k'' = \frac{C'}{C},$$

then the anharmonic ratio of the pencil will have a given value k if

$$(k - k')(k - k'') = 0;$$

that is, if

$$k^2 + k + 1 = 0,$$

or, what is the same thing, if

$$A'^2(2k + 1)^2 + 4BC = 0;$$

that is, if

$$A'^2(2k + 1)^2 - (B - C)^2 = 0,$$

where

$$A'^2 = (a - b)^2(c - d)^2, \quad B - C = (a + b)(c + d) - 2(ab + cd),$$

are each of them symmetrical in regard to a, b , and in regard to c, d , respectively.

2. Let the equations of the two conics be

$$U = (a, b, c, f, g, h \sim x, y, z)^2 = 0, \quad U' = (a', b', c', f', g', h' \sim x, y, z)^2 = 0,$$

and let (α, β, γ) be the coordinates of the variable point.

Putting as usual

$$(A, B, C, F, G, H) = (bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch),$$

$$K = abc - af^2 - bg^2 - ch^2 + 2fgh,$$

the equation of the tangents to the first conic is

$$(A, B, C, F, G, H \sim X, Y, Z)^2 = 0,$$

where

$$X = \beta z - \gamma y, \quad Y = \gamma x - \alpha z, \quad Z = \alpha y - \beta x,$$

and therefore $\alpha X + \beta Y + \gamma Z = 0$.

Hence substituting for Z the value $-\frac{\alpha X + \beta Y}{\gamma}$, we find, for the equation of the tangents, an equation of the form $aX^2 + 2hXY + bY^2 = 0$, which has, in effect, been taken to be $(X - aY)(X - bY) = 0$; that is, we have

$$1 : a + b : ab = a : -2h : b;$$

and, in like manner, if the accented letters refer to the second conic

$$1 : c + d : cd = a' : -2h' : b'.$$

Substituting for a, h, b their values, and for a', h', b' the corresponding values, we find

$$\begin{aligned} 1 : c + d : cd &= A'\gamma^2 - 2G'\gamma\alpha + C'\alpha^2 \\ &: -2(H'\gamma^2 - F'\alpha\gamma - G'\beta\gamma + C'\alpha\beta) \\ &: B'\gamma^2 - 2F'\beta\gamma + C'\beta^2, \\ 1 : a + b : ab &= A\gamma^2 - 2G\gamma\alpha + C\alpha^2 \\ &: -2(H\gamma^2 - F\alpha\gamma - G\beta\gamma + C\alpha\beta) \\ &: B\gamma^2 - 2F\beta\gamma + C\beta^2. \end{aligned}$$

We then have

$$(a - b)^2 = (a + b)^2 - 4ab = -4\gamma^2 K(\alpha, \beta, \gamma)^2,$$

and similarly

$$(c - d)^2 = -4\gamma^2 K'(\alpha, \beta, \gamma)^2.$$

We have, moreover,

$$\begin{aligned} & (a + b)(c + d) - 2(ab + cd) \\ &= 4(H\gamma^2 - F\alpha\gamma - G\beta\gamma + C\alpha\beta)(H'\gamma^2 - F'\alpha\gamma - G'\beta\gamma + C'\alpha\beta) \\ & \quad - 2(B\gamma^2 - 2F\beta\gamma + C\beta^2)(A'\gamma^2 - 2G'\gamma\alpha + C'\alpha^2) \\ & \quad - 2(B'\gamma^2 - 2F'\beta\gamma + C'\beta^2)(A\gamma^2 - 2G\gamma\alpha + C\alpha^2) \\ &= -2\gamma^2(BC' + B'C - 2FF', \dots \sim \alpha, \beta, \gamma)^2, \end{aligned}$$

and substituting the foregoing values, we find

$$4(2k + 1)^2 KK'(\alpha, \dots \sim \alpha, \beta, \gamma)^2 - \{(BC' + B'C - 2FF', \dots \sim \alpha, \beta, \gamma)^2\}^2 = 0,$$

or putting for shortness

$$\Theta = (BC' + B'C - 2FF', GH' + G'H - AF' - A'F, \dots \sim \alpha, \beta, \gamma)^2,$$

the equation of the locus is

$$4(2k + 1)^2 KK' \cdot UU' - \Theta^2 = 0,$$

where (α, β, γ) are current coordinates.

The locus is thus a quartic curve having quadruple contact with each of the conics $U = 0$, $U' = 0$; viz. it touches them at their points of intersection with the conic $\Theta = 0$, which is the locus of the point such that the four tangents form a harmonic pencil.

3. The equation may be written somewhat more elegantly under the form

$$4(2k + 1)^2 \cdot KU \cdot K'U' - \Theta^2 = 0;$$

viz. in this equation we have

$$\begin{aligned} KU &= (BC - F^2, \dots \sim \alpha, \beta, \gamma)^2, \quad K'U' = (B'C' - F'^2, \dots \sim \alpha, \beta, \gamma)^2, \\ &= (BC' + B'C - 2FF', \dots \sim \alpha, \beta, \gamma)^2. \end{aligned}$$

In the last form the equation is expressed in terms of the coefficients $(A, \dots)$, $(A', \dots)$ of the line equations of the conics, viz. these may be taken to be

$$(A, \dots \sim \xi, \eta, \zeta)^2 = 0, \quad (A', \dots \sim \xi, \eta, \zeta)^2 = 0.$$

4. In particular, if each of the conics break up into a pair of points, viz. (l, m, n) and (p, q, r) for the first conic, (l', m', n') and (p', q', r') for the second conic, then the line equations are

$$2(l\xi + m\eta + n\zeta)(p\xi + q\eta + r\zeta) = 0, \quad 2(l'\xi + m'\eta + n'\zeta)(p'\xi + q'\eta + r'\zeta) = 0,$$

so that

$$A = 2lp, \dots F = mr + nq, \dots A' = 2l'p', \dots F' = m'r' + n'q', \dots$$

$(BC - F^2, \dots) = -(mn - nq, np - lr, lq - mp)^2,$
 $(B'C' - F'^2, \dots) = -(m'n' - n'q', n'p' - l'r', l'q' - m'p')^2,$
 $BC' + B'C - 2FF' = 2\{(mn' - m'n)(qr' - q'r) - (nr' - n'r)(m'q - mq'), \dots\},$
 and substituting these values the equation is

$$4(2k+1)^2 \begin{vmatrix} \alpha & \beta & \gamma \\ l & m & n \\ p & q & r \end{vmatrix}^2 \begin{vmatrix} \alpha & \beta & \gamma \\ l' & m' & n' \\ p' & q' & r' \end{vmatrix}^2 - \left[\begin{vmatrix} \alpha & \beta & \gamma \\ l & m & n \\ p & q & r \end{vmatrix} \begin{vmatrix} \alpha & \beta & \gamma \\ l' & m' & n' \\ p' & q' & r' \end{vmatrix} \right]^2 = 0,$$

which, if A, B, C denote

$$\begin{vmatrix} \alpha & \beta & \gamma \\ l & m & n \\ p & q & r \end{vmatrix}, \quad \begin{vmatrix} \alpha & \beta & \gamma \\ l' & m' & n' \\ p & q & r \end{vmatrix}, \quad \begin{vmatrix} \alpha & \beta & \gamma \\ l & m & n \\ p' & q' & r' \end{vmatrix} \text{ respectively, } (A + B + C = 0)$$

is, in fact, the equation

$$(2k+1)^2 A^2 - (B - C)^2 = 0,$$

that is

$$k = -\frac{B}{A} \text{ or } k = -\frac{C}{A},$$

either of which expresses the anharmonic property of the points of a conic in the form given by the theorem *ad quatuor lineas*.

5. Reverting to the case of two conics, then if these be referred to a set of conjugate axes, the equations will be

$$aa^2x^2 + bb^2y^2 + cc^2z^2 = 0, \quad a'a'^2x^2 + b'b'^2y^2 + c'c'^2z^2 = 0,$$

we have $K = abc, K' = a'b'c'$,

$$\Theta = (bc' + b'c)aa'\alpha^2 + (ca' + c'a)bb'\beta^2 + (ab' + a'b)cc'\gamma^2,$$

and the equation of the quartic curve is

$$4(2k+1)^2 abca'b'c'(aa^2x^2 + bb^2y^2 + cc^2z^2)(a'a'^2x^2 + b'b'^2y^2 + c'c'^2z^2) - \Theta^2 = 0.$$

6. I suppose in particular that the two conics are

$$x^2 + my^2 - 1 = 0, \quad mx^2 + y^2 - 1 = 0,$$

the equation of the quartic is

$$4(2k+1)^2 m^2 (x^2 + my^2 - 1)(mx^2 + y^2 - 1) - \{(m^2 + m)(x^2 + y^2) - m^2 - 1\}^2 = 0;$$

or putting $\lambda = \frac{(m+1)^2}{(2k+1)^2}$, this is

$$(x^2 + y^2 - \lambda)^2 - \lambda^2 (x^2 + my^2 - 1)(mx^2 + y^2 - 1) = 0.$$

To fix the ideas, suppose that m is positive and > 1 , so that each of the conics is an ellipse, the major semi-axis being $= 1$, and the minor semi-axis being $= \frac{1}{\sqrt{m}}$.

For any real value of k the coefficient λ is positive, and it may accordingly be assumed that λ is positive.

We have $\frac{m^2 + 1}{m(m + 1)} < \lambda < 1$, or the radius of the circle is intermediate between the semi-axes of the ellipses, hence the points of contact on each ellipse are real points.

Writing for shortness $\alpha = \frac{m^2 + 1}{m^2 + \lambda}$, the equation is

$$(x^2 + my^2 - 1)(mx^2 + y^2 - 1) - \lambda(x^2 + y^2 - \alpha)^2 = 0.$$

For the points on the axis of x , we have

$$(x^2 - 1)(mx^2 - 1) - \lambda(x^2 - \alpha)^2 = 0,$$

that is

$$(m - \lambda)x^4 + \{-(1 + m) + 2\lambda\alpha\}x^2 + (1 - \lambda\alpha^2) = 0,$$

and thence

$$(m - \lambda)x^2 = \frac{1}{2}(1 + m) - \lambda\alpha \pm \frac{1}{2}\sqrt{\{(m - 1)^2 + 4\lambda(1 - \alpha)(1 - m\alpha)\}},$$

or, substituting for α its value, this is

$$x^2 = \frac{\lambda(m + \frac{1}{m})}{\lambda^2 - m}, \dots$$

The investigation shows that for different values of k , the locus consists of a series of quartic curves, each having quadruple contact with the two given conics, and touching them at points on the conic $\Theta = 0$.

390. Theorem relating to the four Conics which touch the same two lines and pass through the same four points

[From the Quarterly Journal of Pure and Applied Mathematics, vol. VIII. (1867), pp. 162-167.]

1. The theorem referred to in the title is as follows:

Theorem 2. *If four conics touch the same two lines and pass through the same four points, then the four poles of the line joining the points of contact of the two tangents, in regard to the four conics respectively, lie on a right line.*

The proof is very simple. Let the two lines be taken for the lines $x = 0, y = 0$; and let the four points be the intersections of the lines $z = 0$ with the conic $(a, b, c, f, g, h \sim x, y, z)^2 = 0$.

The equation of a conic touching the lines $x = 0, y = 0$, and passing through the four points, is

$$(a, b, c, f, g, h \sim x, y, z)^2 + 2\lambda xy = 0,$$

where λ is an arbitrary parameter. This conic touches the two lines at the points where the line $fx + gy + hz = 0$ meets them; that is, at the points $(0, -h, g)$ and $(-h, 0, f)$.

The pole of the line joining these two points, that is, of the line $fgx + fhy - h^2z = 0$, in regard to the conic, is found to be

$$x : y : z = -h(g + \lambda f) : -h(f + \lambda g) : fg - \lambda h^2.$$

For the four values of λ corresponding to the four conics, we have four such poles; and it is easy to verify that these four poles lie on the line

$$fgx - fgy + (f^2 - g^2)z = 0.$$

2. The theorem may be stated in a different form. If four conics pass through the same four points, their eight tangents at these points form an interesting configuration. The theorem then asserts that the poles of the line joining the points of contact of any two fixed tangents, with respect to the four conics, are collinear.
3. The theorem is a particular case of a more general theorem relating to curves of any order. If n curves of the order m pass through the same mn points, and touch the same m lines, then the poles of any one of these lines, in regard to the n curves, lie on a curve of order $n - 1$.
4. In the case of conics, the number of conics which touch two given lines and pass through four given points is $= 4$. For the conic is determined by 5 conditions; the conditions of touching two lines are 2 conditions, and passing through 4 points is 4 conditions, together 6 conditions; but there is one condition too many, since the 4 points and 2 lines form a system of 6 elements which must satisfy a condition in order that conics touching the 2 lines and passing through the 4 points may exist.
5. The condition in question may be found as follows. Let the four points be the intersections of a conic $U = 0$ with a line $z = 0$; and let the two lines be $x = 0, y = 0$. Then if a conic $V = 0$ touches the lines $x = 0, y = 0$ and passes through the four points, we must have

$$V = U + 2\lambda xy = 0,$$

and the condition that this conic touches $x = 0$ and $y = 0$ gives two equations for λ , which in general have 4 common solutions.

6. The theorem may be connected with the theory of the invariants of two conics. If $U = 0, V = 0$ are two conics, the condition that the conic $U + \lambda V = 0$ may touch a given line gives an equation in λ , the roots of which determine the conics of the pencil which touch the given line. In the present case, the pencil is $U + 2\lambda xy = 0$, and the condition of touching $x = 0$ and $y = 0$ leads to a quartic equation in λ , giving the 4 conics of the theorem.
7. The theorem admits of an interesting geometrical interpretation. Consider the four conics as the conics of a pencil which touch two fixed lines. The poles of the line joining the points of contact form a range of four points on a line; and this range has properties analogous to those of the range of four points in which the conics of a pencil meet a fixed line.
8. I remark in conclusion that the theorem is intimately connected with the theory of the bitangents of a quartic curve. If we consider the quartic curve enveloped by the tangents to the four conics, the line of poles is related to the configuration of the bitangents in a remarkable manner.

391. Solution of a Problem of Elimination

[From the Quarterly Journal of Pure and Applied Mathematics, vol. VIII. (1867), pp. 183–185.]

1. The problem of elimination which I propose to solve is the following:

Given the system of equations

$$\begin{aligned} ax^2 + 2hxy + by^2 + 2gxz + 2fyz + cz^2 &= 0, \\ a'x^2 + 2h'xy + b'y^2 + 2g'xz + 2f'yz + c'z^2 &= 0, \\ a''x^2 + 2h''xy + b''y^2 + 2g''xz + 2f''yz + c''z^2 &= 0, \end{aligned}$$

representing three conics; to find the condition that the three conics may have a common point.

2. The required condition is obtained by eliminating (x, y, z) from the three equations. The most symmetrical form of the result is as a determinant. If we write the equations in the form

$$U = 0, \quad U' = 0, \quad U'' = 0,$$

then the condition is that the determinant

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} \begin{vmatrix} a' & h' & g' \\ h' & b' & f' \\ g' & f' & c' \end{vmatrix} \begin{vmatrix} a'' & h'' & g'' \\ h'' & b'' & f'' \\ g'' & f'' & c'' \end{vmatrix} = 0,$$

where the product is formed in a certain symbolic manner.

3. The actual expression of the condition of elimination may be presented under a very elegant form by the use of the symbolic notation for determinants. If the three conics are written as

$$(a, b, c, d, e, f \sim x, y, z)^2 = 0, \quad (a', \dots \sim x, y, z)^2 = 0, \quad (a'', \dots \sim x, y, z)^2 = 0,$$

then the result of elimination is:

Represent the columns

$$\begin{array}{cccccc} a & b & a' & b' & a'' & b'' \\ b & c & b' & c' & b'' & c'' \\ c & d & c' & d' & c'' & d'' \\ d & e & d' & e' & d'' & e'' \\ e & f & e' & f' & e'' & f'' \end{array}$$

by 1, 2, 3, 4, 5, 6, then each equation is of the type

$$\lambda(1 + k \cdot 2) + \lambda'(3 + k \cdot 4) + \lambda''(5 + k \cdot 6) = 0.$$

Multiplying the several equations by the minors of 135, each with its proper sign, and adding, the terms independent of k disappear, the equation divides by k , and we find

$$\lambda \cdot 2135 + \lambda' \cdot 4135 + \lambda'' \cdot 6135 = 0;$$

operating in a similar manner with the minors of 246, the terms in k disappear, and we find

$$\lambda \cdot 1246 + \lambda' \cdot 3246 + \lambda'' \cdot 5246 = 0;$$

again, operating with the minors of $(146 + 236 + 245 + k \cdot 246)$, we find

$$\begin{aligned} & \lambda\{1236 + 1245 + k(2146 + 1246)\} \\ & + \lambda'\{3146 + 3245 + k(4236 + 3246)\} \\ & + \lambda''\{5146 + 5236 + k(6245 + 5246)\} = 0, \end{aligned}$$

where the terms in k disappear, and this is

$$\lambda(1236 + 1245) + \lambda'(3146 + 3245) + \lambda''(5146 + 5236) = 0.$$

We have thus three linear equations, which written in a slightly different form are

$$\begin{aligned} & \lambda \cdot 1235 + \lambda' \cdot 3451 + \lambda'' \cdot 5613 = 0, \\ & \lambda(1236 + 1245) + \lambda'(3452 + 3461) + \lambda''(5614 + 5623) = 0, \\ & \lambda \cdot 1246 + \lambda' \cdot 3462 + \lambda'' \cdot 5624 = 0, \end{aligned}$$

and thence eliminating $\lambda, \lambda', \lambda''$, we have

$$\begin{vmatrix} 1235, & 1236 + 1245, & 1246 \\ 3451, & 3452 + 3461, & 3462 \\ 5613, & 5614 + 5623, & 5624 \end{vmatrix} = 0,$$

which is the required result.

4. It may be remarked that the second and third column are obtained from the first by operating on it with $\Delta, \frac{1}{2}\Delta^2$, if $\Delta = 23_1 + 45_3 + 65_5$.

Or say the result is

$$(1, \Delta, \tfrac{1}{2}\Delta^2) \begin{vmatrix} 1235 \\ 3451 \\ 5613 \end{vmatrix} = 0.$$

In like manner for the system

$$\begin{array}{cccccc} x^2 & xy & y^2 & yz & z^2 & zx \\ a & b & c & d & e & f \\ a' & b' & c' & d' & e' & f' \\ a'' & b'' & c'' & d'' & e'' & f'' \\ a''' & b''' & c''' & d''' & e''' & f''' \end{array}$$

if the columns are

$$\begin{array}{cccccc} a & b, & a' & b', & a'' & b'', & a''' & b''' \\ b & c, & b' & c', & b'' & c'', & b''' & c''' \\ c & d, & c' & d', & c'' & d'', & c''' & d''' \\ d & e, & d' & e', & d'' & e'', & d''' & e''' \\ e & f, & e' & f', & e'' & f'', & e''' & f''' \end{array}$$

then the result is

$$(1, \Delta, \tfrac{1}{2}\Delta^2, \tfrac{1}{6}\Delta^3) \begin{vmatrix} 12357 \\ 34571 \\ 56713 \\ 78135 \end{vmatrix} = 0,$$

where

$$\Delta = 28_1 + 48_3 + 68_5 + 8\chi_7.$$

392. On the Conics which pass through two given Points and touch two given Lines

[From the Quarterly Journal of Pure and Applied Mathematics, vol. VIII. (1867), pp. 211-219.]

1. Let $x = 0$, $y = 0$ be the equations of the given lines; $z = 0$ the equation of the line joining the given points. We may, to fix the ideas, imagine the implicit constants so determined that $x + y + z = 0$ shall be the equation of the line infinity.

Take $x - my = 0$, $x - ny = 0$ as the equations of the lines which by their intersection with $z = 0$ determine the given points.

The equation of the conic is

$$\sqrt{xy(m)} + \sqrt{xy(n)} \cdot \sqrt{xy} = x + y\sqrt{mn} + \gamma z,$$

or, what is the same thing,

$$(x - my)(x - ny) + 2(x + y\sqrt{mn})\gamma z + \gamma^2 z^2 = 0,$$

so that there are two distinct series of conics according as $\sqrt{mn}$ is taken with the positive or the negative sign.

The equation of the chord of contact is

$$x + y\sqrt{mn} + \gamma z = 0,$$

which meets $z = 0$ in the point

$$[x + y\sqrt{mn} = 0, \quad z = 0]$$

that is in one of the centres of the involution formed by the lines $(x = 0, y = 0)$, $(x - my = 0, x - ny = 0)$.

It is to be observed that the conic is only real when mn is positive, that is (the lines and points being each real) the two points must be situate in the same region or in opposite regions of the four regions formed by the two lines: there are however other real cases; e.g. if the lines $x = 0$, $y = 0$ are real, but the quantities m , n are conjugate imaginaries; included in this we have the circles which touch two real lines.

2. To fix the ideas I take m and n each positive and $mn > 1$; also I attend first to the series where $\sqrt{mn}$ is taken positively.

At the points where the conic meets infinity, we have

$$(\sqrt{m} + \sqrt{n})\sqrt{xy} = x + y\sqrt{mn} - \gamma(x + y),$$

which gives two coincident points, that is the conic is a parabola, if

$$(1 - \gamma)\{\sqrt{mn} - \gamma\} = \frac{1}{4}(\sqrt{m} + \sqrt{n})^2,$$

that is

$$\gamma = -\frac{1}{2}(1 + \sqrt{mn}) \pm \frac{1}{2}\sqrt{(1 + m)(1 + n)},$$

or

$$\gamma = \frac{1}{2}[1 + \sqrt{mn} + \sqrt{(1 + m)(1 + n)}],$$

where it is to be noticed that

$$\gamma = \frac{1}{2}[1 + \sqrt{mn} + \sqrt{(1+m)(1+n)}]$$

is a positive quantity greater than $\sqrt{mn}$, say $\gamma = p$;

$$\gamma = \frac{1}{2}[1 + \sqrt{mn} - \sqrt{(1+m)(1+n)}]$$

is a negative quantity, say $\gamma = -q$, q being positive.

The order of the lines is as shown in fig. 1, see plate facing p. 52.

$\gamma = -\infty$ to $\gamma = -q$,	curve is ellipse;
$\gamma = -q$,	parabola P_1 ,
$\gamma = -q$ to p ,	curve is hyperbola;
$\gamma = p$,	parabola P_2 ,
$\gamma = p$ to $\gamma = \infty$,	ellipse.

3. Resuming the equation

$$(x - my)(x - ny) + 2(x + y\sqrt{mn})\gamma z + \gamma^2 z^2 = 0,$$

the coefficients are

$$(a, b, c, f, g, h) = (1, mn, \gamma^2, \gamma\sqrt{mn}, \gamma, -\frac{1}{2}(m+n)),$$

and thence the inverse coefficients are

$$\begin{aligned} (A, B, C, F, G, H) = & (0, 0, -\frac{1}{4}(m-n)^2, \\ & -\frac{1}{4}\gamma(\sqrt{m} + \sqrt{n})^2, \\ & -\frac{1}{2}\gamma\sqrt{mn}(\sqrt{m} + \sqrt{n})^2, \\ & \frac{1}{4}\gamma^2(\sqrt{m} + \sqrt{n})^2). \end{aligned}$$

$$K = -\frac{1}{4}\gamma^2(\sqrt{m} + \sqrt{n})^2,$$

or, omitting a factor, the inverse coefficients are

$$(A, B, C, F, G, H) = (0, -\gamma(\sqrt{m} - \sqrt{n})^2, 1, \frac{1}{\gamma}\sqrt{mn}, -\gamma, \dots).$$

Considering the line $\lambda x + \mu y + \nu z = 0$, the coordinates of the pole of this line are

$$X : Y : Z = -\gamma\mu + \sqrt{mn}\nu : -\gamma\lambda + \mu : \sqrt{mn}\lambda + \mu + \frac{1}{\gamma}(\sqrt{m} - \sqrt{n})^2\nu.$$

4. In particular, if $\lambda = \mu = \nu = 1$, then for the coordinates of the centre of the conic, we have

$$X : Y : Z = -\gamma + \sqrt{mn} : -\gamma + 1 : \sqrt{mn} + 1 + \frac{1}{\gamma}(\sqrt{m} - \sqrt{n})^2;$$

and for the locus of the centre,

$$(X - Y)^2 \cdot \frac{1}{4}(\sqrt{m} - \sqrt{n})^2 + (X - Y)(X - Y\sqrt{mn})\{1 + \sqrt{mn}\} + z(X - Y\sqrt{mn})\{1 - \sqrt{mn}\} = 0,$$

so that the locus is a conic, and it is obvious that this conic is a hyperbola.

Putting for greater simplicity

$$X = x - y, \quad Y = x - y\sqrt{mn}, \quad z = Z,$$

the equation of the curve of centres is

$$ZQ + X^2 \cdot \frac{1}{4}(\sqrt{m} - \sqrt{n})^2 + XY\{1 + \sqrt{mn}\} + YZ\{1 - \sqrt{mn}\} = 0,$$

or, writing this under the form

$$Z[Y\{1 + \sqrt{mn}\} + Z\{1 - \sqrt{mn}\}] + \frac{1}{4}(\sqrt{m} - \sqrt{n})^2 X^2 = 0,$$

the equation is

$$ZQ + XY' = 0,$$

where

$$X = x - y, \quad Y = x - y\sqrt{mn}, \quad Z = \frac{2}{(\sqrt{m} - \sqrt{n})^2} [Y' - (x - y) + \{1 - \sqrt{mn}\}z];$$

these values give

$$\begin{aligned} x - y &= X, & x - y\sqrt{mn} &= Y, \\ \{1 - \sqrt{mn}\}z &= \frac{1}{2}(\sqrt{m} - \sqrt{n})^2 Q + 2\{1 + \sqrt{mn}\}X, \end{aligned}$$

or, what is the same thing,

$$\begin{aligned} \{1 - \sqrt{mn}\}x &= -\sqrt{mn}X + Y, \\ \{1 - \sqrt{mn}\}y &= -X + Y, \\ \{1 - \sqrt{mn}\}z &= 2\{1 + \sqrt{mn}\}X + \frac{1}{2}(\sqrt{m} - \sqrt{n})^2 Q. \end{aligned}$$

whence also

$$\{1 - \sqrt{mn}\}(x + y + z) = \{1 + \sqrt{mn}\}Z + 2Y + (\sqrt{m} - \sqrt{n})^2 Q,$$

or the equation of the line infinity is

$$\{1 + \sqrt{mn}\}X + 2Y + (\sqrt{m} - \sqrt{n})^2 Q = 0,$$

a formula which may be applied to finding the asymptotes and thence the centre of the conic $QZ + X^2 = 0$.

In fact we have identically

$$[2kx + 2ky - (2k + 1)z]^2 - (1 + 4k)(2kx - z)^2 = 4k^2(x + y + z)^2 - 4k(1 + 4k)(kx^2 + yz),$$

that is

$$-4k(1 + 4k)(kx^2 + yz) = [2kx + 2ky - (2k + 1)z]^2 - (1 + 4k)(2kx - z)^2 - 4k^2(x + y + z)^2,$$

which, if $x + y + z = 0$ is the equation of the line infinity, puts in evidence the asymptotes of the conic $kx^2 + yz = 0$.

Hence writing αx , βy , γz in the place of x , y , z respectively and $k = k'$, that is, $k = \frac{k'}{\alpha^2}$, we have

$$-4\beta^2\gamma k'(\alpha^2 + 4\beta\gamma k')(k'x^2 + yz) = [2\alpha\gamma k'x + 2\beta^2\gamma k'y - (2\beta^2\gamma k' + \alpha^2)\gamma z]^2 - (\alpha^2 + 4\beta^2\gamma k')[2\beta\gamma k'x - \alpha z]^2 - 4\beta^2\gamma^2 k'^2(\alpha x + \beta y + \gamma z)^2,$$

or, what is the same thing,

$$\begin{aligned} -4\beta^2\gamma k'(\alpha^2 + 4\beta\gamma k')(k'x^2 + yz) \\ = [2\alpha\gamma k'x + 2\beta^2\gamma k'y - (2\beta^2\gamma k' + \alpha^2)\gamma z]^2 \\ - (\alpha^2 + 4\beta^2\gamma k')(2\beta\gamma k'x - \alpha z)^2 - 4\beta^2\gamma^2 k'^2(\alpha x + \beta y + \gamma z)^2, \end{aligned}$$

which, when $\alpha x + \beta y + \gamma z = 0$ is the equation of the line infinity, puts in evidence the asymptotes of the conic $k'x^2 + yz = 0$.

Now writing X, Y, Q in the place of x, y, z ; $k' = 1$, and $\alpha = \{1 + \sqrt{mn}\}$, $\beta = 2$, $\gamma = (\sqrt{m} - \sqrt{n})^2$, we have

$$\begin{aligned} & -16[\{1 + \sqrt{mn}\}^2 + 8(\sqrt{m} - \sqrt{n})^2](ZQ + X^2) \\ & = [4\{1 + \sqrt{mn}\}Z + 8Y - (4(\sqrt{m} - \sqrt{n})^2 + \{1 + \sqrt{mn}\}^2)Q]^2 \\ & \quad - [\{1 + \sqrt{mn}\}^2 + 8(\sqrt{m} - \sqrt{n})^2][4X - \{1 + \sqrt{mn}\}Q]^2 \\ & \quad - 16[\{1 + \sqrt{mn}\}X + 2Y + (\sqrt{m} - \sqrt{n})^2Q]^2, \end{aligned}$$

and the asymptotes are

$$\begin{aligned} & 4\{1 + \sqrt{mn}\}Z + 8Y - [4(\sqrt{m} - \sqrt{n})^2 + \{1 + \sqrt{mn}\}^2]Q \\ & = \pm \sqrt{\{1 + \sqrt{mn}\}^2 + 8(\sqrt{m} - \sqrt{n})^2} [4X - \{1 + \sqrt{mn}\}Q]. \end{aligned}$$

At the centre

$$\begin{aligned} & 4\{1 + \sqrt{mn}\}Z + 8Y - [4(\sqrt{m} - \sqrt{n})^2 + \{1 + \sqrt{mn}\}^2]Q = 0, \\ & 4X - \{1 + \sqrt{mn}\}Q = 0, \end{aligned}$$

but the first equation is

$$\{1 + \sqrt{mn}\}[4Z - Q\{1 + \sqrt{mn}\}] + 8Y - 4(\sqrt{m} - \sqrt{n})^2Q = 0,$$

so that we have

$$4Z = \{1 + \sqrt{mn}\}Q, \quad 2Y = (\sqrt{m} - \sqrt{n})^2Q,$$

the first of these is

$$2(\sqrt{m} - \sqrt{n})^2(x - y) - \{1 + \sqrt{mn}\}^2(x - y) - (1 - mn)z = 0,$$

and the two together give

$$2Z(\sqrt{m} - \sqrt{n})^2 - \{1 + \sqrt{mn}\}Y = 0,$$

so that we have

$$\begin{aligned} & 2(\sqrt{m} - \sqrt{n})^2(x - y) - \{1 + \sqrt{mn}\}\{x - y\sqrt{mn}\} = 0, \\ & [\{1 + \sqrt{mn}\}^2 - 2(\sqrt{m} - \sqrt{n})^2](x - y) + (1 - mn)z = 0, \end{aligned}$$

to determine the coordinates of the centre.

5. The equation of the chord of contact is

$$x + y\sqrt{mn} + \gamma z = 0,$$

which for $\gamma = 1$ is parallel to $y = 0$ and for $\gamma = \sqrt{mn}$ is parallel to $x = 0$.

But the coordinates of the centre are

$$X : Y : Z = -\gamma + \sqrt{mn} : -\gamma + 1 : \sqrt{mn} + 1 + \frac{1}{\gamma}(\sqrt{m} - \sqrt{n})^2,$$

which for $\gamma = 1$ give

$$y = 0, \quad X : z = -1 + \sqrt{mn} : \sqrt{mn} + 1 + \frac{1}{4}(\sqrt{m} - \sqrt{n})^2 = -2 + 2\sqrt{mn} : 2 + m + n,$$

and for $\gamma = \sqrt{mn}$ give

$$x = 0, \quad Y : z = 1 - \sqrt{mn} : \sqrt{mn} + 1 + \frac{1}{4\sqrt{mn}}(\sqrt{m} - \sqrt{n})^2 = 2 - 2\sqrt{mn} : 2\sqrt{mn} + \frac{(m-n)^2}{4\sqrt{mn}}.$$

The line drawn from the fixed point on the chord of contact to the centre has for its equation

$$x + y\sqrt{mn} + \gamma'z = 0,$$

where, writing for x, y, z the coordinates of the centre, we have

$$-\gamma\{1 + \sqrt{mn}\} + 2\sqrt{mn} + \gamma'\{\sqrt{mn} + 1 + \frac{1}{4}(\sqrt{m} - \sqrt{n})^2\} = 0,$$

that is

$$\gamma' = \frac{\gamma\{1 + \sqrt{mn}\} - 2\sqrt{mn}}{\sqrt{mn} + 1 + \frac{1}{4}(\sqrt{m} - \sqrt{n})^2}.$$

6. I remark that the four conics which pass through the two given points and touch the two given lines form a system of considerable interest. The two series of conics (corresponding to the two signs of $\sqrt{mn}$) are distinct, and each series contains an infinite number of conics. The parabolic conics in each series correspond to particular values of the parameter γ , and the locus of centres is in each case a conic.

The theorem that the locus of the centres is a conic is a particular case of a more general theorem relating to the locus of the poles of a given line in regard to a system of conics passing through fixed points and touching fixed lines.